Food Waste Prediction

1. **How can Machine Learning algorithms predict food waste using historical consumption, customer demand, weather, and inventory data?**

   To identify relationships between factors such as customer demand, weather, and the amount of food wasted**.**

   Statistical Test :Correlation TesT

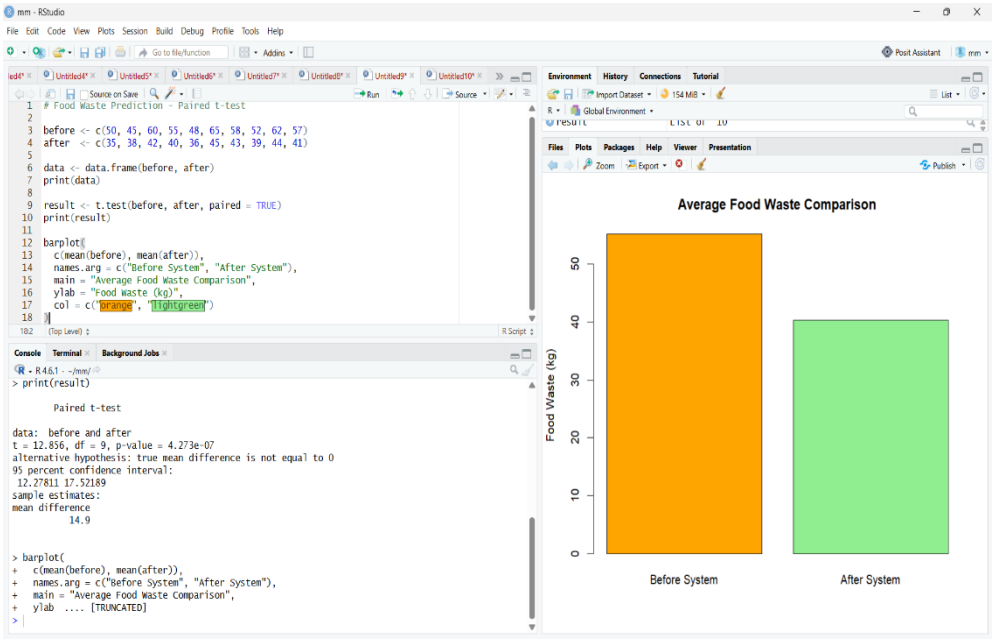
   Formula: Pearson Correlation Coefficient

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

   Purpose: To identify whether higher customer demand is associated with higher food waste.

   CODE: # Food Waste Prediction - Correlation Test

```
demand <- c(100, 120, 150, 180, 200, 220, 250, 280, 300, 320)
waste <- c(10, 15, 18, 20, 25, 28, 30, 35, 38, 40)
data <- data.frame(demand, waste)
print(data)
result <- cor.test(demand, waste, method = "pearson")
print(result)
plot(demand, waste,
   main = "Customer Demand vs Food Waste",
   xlab = "Customer Demand",
   ylab = "Food Waste (kg)",
   pch = 19)
abline(lm(waste ~ demand), col = "red")
```



2. **How can a Food Waste Prediction system help restaurants, supermarkets, and households reduce food wastage and costs?**

Statistical Test :t-test

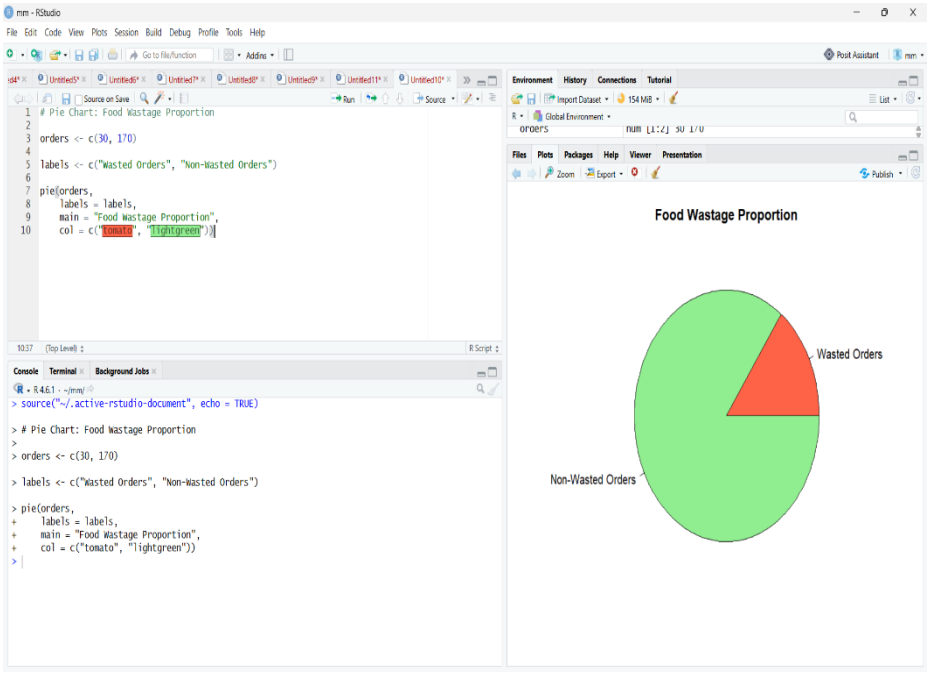
Formula:$t=sd/n\bar{d}$

Visualization: Bar Chart

- X-axis: Before System and After System
- Y-axis: Average Food Waste (kg)
- Two bars compare the average waste before and after implementing the system.

Purpose: To determine whether average food waste changes significantly after implementing the system.

CODE:
```
# Food Waste Prediction - Paired t-test
before <- c(50, 45, 60, 55, 48, 65, 58, 52, 62, 57)
after  <- c(35, 38, 42, 40, 36, 45, 43, 39, 44, 41)
data <- data.frame(before, after)
print(data)
result <- t.test(before, after, paired = TRUE)
print(result)
barplot(
 c(mean(before), mean(after)),
 names.arg = c("Before System", "After System"),
 main = "Average Food Waste Comparison",
 ylab = "Food Waste (kg)",
 col = c("orange", "lightgreen")
)
```



3. **Is the proportion of food orders wasted significantly different from the expected wastage rate of 10%?**

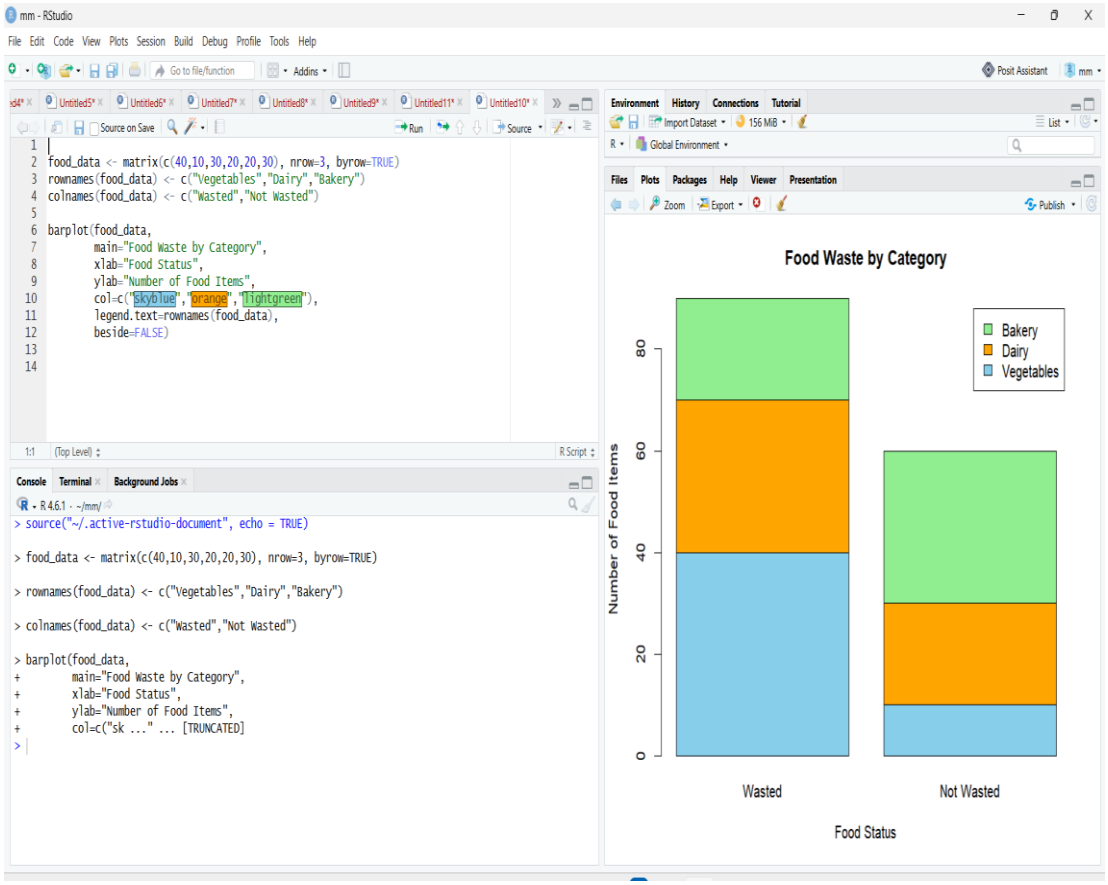
Statistical Test: One-sample Proportion Test

Formula:

$$\hat{p} = \frac{x}{n}$$

CODE:

```
orders <- c(30, 170)
labels <- c("Wasted Orders", "Non-Wasted Orders")
pie(orders,
  labels = labels,
  main = "Food Wastage Proportion",
  col = c("tomato", "lightgreen"))
```



4.Is there a significant association between food category and whether food is wasted?

Statistical Test: Chi-square Test of Independence

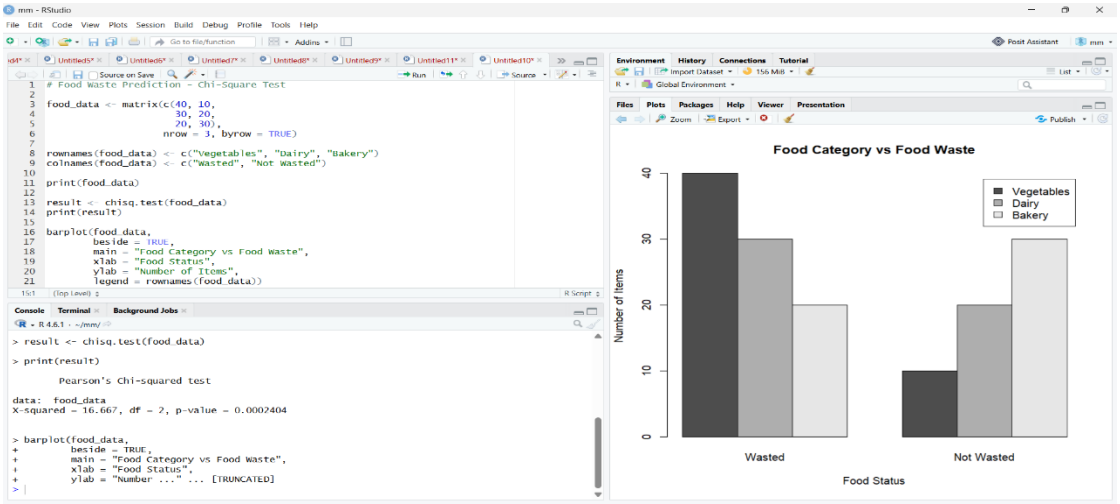
Formula:

$\chi 2=\sum E(O-E)2$

Purpose: To determine whether food category and waste status are statistically associated.

CODE:

```
# Food Waste Prediction - Chi-Square Test
food_data <- matrix(c(40, 10,
                      30, 20,
                      20, 30),
                    nrow = 3, byrow = TRUE)
rownames(food_data) <- c("Vegetables", "Dairy", "Bakery")
colnames(food_data) <- c("Wasted", "Not Wasted")
print(food_data)
result <- chisq.test(food_data)
print(result)
barplot(food_data,
        beside = TRUE,
        main = "Food Category vs Food Waste",
        xlab = "Food Status",
        ylab = "Number of Items",
        legend = rownames(food_data))
```



5. Does average food waste differ among restaurants, supermarkets, and households?

Statistical Test: One-way ANOVA

Formula: F=MSW/MSB

CODE:

```
restaurants <- c(40, 45, 42, 38, 44)
supermarkets <- c(60, 65, 62, 58, 64)
households <- c(20, 25, 22, 18, 24)
avg_waste <- c(mean(restaurants), mean(supermarkets), mean(households))
groups <- c("Restaurant", "Supermarket", "Household")
plot(avg_waste, type="o", xaxt="n",
    main="Average Food Waste by Group",
    xlab="Group", ylab="Average Food Waste (kg)",
    pch=19)
axis(1, at=1:3, labels=groups)
```

